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KHWOPA COLLEGE OF ENGINEERING

AFFILIATED TO TRIBHUVAN UNIVERSITY

A
Report on

Illumination design of a Commercial Complex

(As part of the project requirements for the course on Industrial Power Distribution and Illumination in the Bachelor's Degree in Electrical Engineering)
(Course Code: EE653)

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ABSTRACT

This project focuses on the illumination and power distribution system for a commercial building that includes coffee shops, a restaurant, and office spaces. The design integrates both lighting and power sub-circuits to ensure efficient energy management and a reliable electrical supply. The lighting system is tailored to create an inviting ambiance for dining areas and a productive environment for offices while maintaining energy efficiency. The power distribution plan ensures proper load balancing, safety compliance, and optimal performance of electrical equipment. Energy-efficient solutions, such as LED lighting and sustainable power management strategies, are implemented to enhance operational efficiency and reduce costs. The project prioritizes safety, user comfort, and regulatory adherence, aiming to deliver a well-structured and sustainable electrical system for the building.

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1 Introduction

This project focuses on the essential role of effective illumination and industrial power distribution in optimizing the functionality, safety, and energy efficiency of commercial buildings. It emphasizes the importance of advanced design and implementation of lighting systems and electrical networks to meet the varied needs of residents while complying with energy efficiency standards. The goal is to create a well-rounded plan that enhances the building's operational capabilities through innovative lighting and power systems.

In the context of electrical engineering, industrial power distribution and illumination entail several key tasks, including design, estimation, control, protection, and ongoing maintenance of lighting and power systems. These systems are specifically tailored for the unique requirements of industrial and commercial environments. Unlike residential setups, industrial power systems demand precise planning and design to optimize costs while ensuring efficient and reliable performance.

This project highlights the significance of well-engineered power systems, which not only minimize electricity costs over time but also maintain stable voltage regulation under challenging conditions. This can be achieved by identifying the load centers of the areas and determining optimal locations for placing the main distribution boards or transformers. A properly designed illumination system will deliver the required lighting levels, enhance safety, improve energy efficiency, and include provisions for emergency backup power.

2 Description of Building

The commercial complex of this project has a total of 7 floors(counting the basement floor too). The brief description of each of the floors is mentioned as below:

2.1 Basement Floor

The basement floor of the commercial building comprises parking facilities for both motorcycles and cars, along with a designated guard room and a storage area. The total area requiring illumination in this section has been calculated as 831.2269 m^2.

2.2 Ground Floor

The ground floor features a combination of commercial establishments and refreshment outlets, including a coffee shop and a retail store specializing in mango products. The total area to be illuminated in the ground floor is 590.524 m^2.

2.3 First Floor

The first floor consists of office spaces, small meeting rooms, and a storage room. Additionally, there are three attached washrooms to accommodate occupants. The total area to be illuminated on the first floor is 449.995329 m^2.

2.4 Second Floor

The second floor includes two office spaces, a lobby, a corridor designated for emergency evacuation, and washroom facilities. The total area to be illuminated on the second floor is 411.4467 m^2.

2.5 Third to Fifth Floor Plan

The third, fourth, and fifth floors follow a similar structural layout. Each floor contains three office spaces and four washrooms, with two washrooms allocated per office. The total area to be illuminated on the floor(Third to fifth) is 553.8628m^2.

2.6 Top Floor Plan

The topmost floor houses a large office space along with two washrooms. The total area to be illuminated on the top floor is 290.7090 m^2. Here are the most common rooms present on each floor of the building:

- Lobby (245.581 m^2)
- Emergency Exit Corridor (417.8441 m^2)
- Electrical Room (23.4113 m^2)

It is important to note that the basement floor does not include the lobby or the emergency exit corridor for obvious reasons. Additionally, the area values recorded in the spreadsheet for the above 3 rooms may vary slightly due to marginal errors made by the designer when calculating the area for each floor.

3 METHODS OF CALCULATING THE NUMBER OF LUMINARIES

There are several methods available for determining the required number of luminaries to illuminate a given area. A few of these methods are mentioned below:

3.1 Lumen Method (Light Flux Method)

- This method is widely used for general lighting calculations.
- It is based on the total luminous flux (lumens) required for a given area and the lumen output of each fixture.

The number of luminaires can be determined using the formula:

$$N = \frac{E \times A}{\phi \times \mu_m \times \mu_u} \tag{1}$$

where:

- N = Number of luminaires

- E = Required illuminance (lux)
- A = Area to be illuminated (m^2)
- ϕ = Lamp flux in lumens
- μ_m = Maintainance factor=$\frac{Illumination\ under\ normal\ working\ conditions}{Illumination\ when\ everything\ is\ perfectly\ clean}$
- μ_u = Utilization factor=$\frac{Total\ lumens\ reaching\ the\ working\ plane}{Total\ lumen\ given\ out\ by\ the\ lamp}$

3.2 Point-by-Point Method

- This method is used for precise lighting calculations, especially in spaces requiring uniform illumination.
- It involves calculating the illuminance at specific points using the inverse square law and cosine law.

$$E = \frac{I}{d^2} \times \cos\theta \tag{2}$$

where:

- E = Illuminance at a point (lux)
- I = Luminous intensity of the source (candela)
- d = Distance from the light source to the point (m)
- θ = Angle of incidence

3.3 Watts per Square Meter Method

- A simple approximation method or rule of thumb method used in lighting designs.
- It assumes a standard wattage per square meter based on the type of space (e.g., office, warehouse, residential).

The required number of luminaires is given by:

$$N = \frac{P \times A}{W} \tag{3}$$

where:

- P = Recommended power density (W/m^2)
- A = Area to be illuminated (m^2)
- W = Wattage of a single fixture (W)

3.4 Room Index Method (Coefficient of Utilization Method)

- This method is commonly used for indoor lighting calculations.
- The room index (RI) is calculated based on the dimensions of the room.

$$RI = \frac{L \times W}{H_m(L + W)} \tag{4}$$

where:

- L = Room length (m)
- W = Room width (m)
- H_m = Mounting height of the luminaire (m)

Each of these methods is chosen based on the level of precision required and the type of space being illuminated.
The method that we used to illuminate the building is the lumen method.

3.5 Selection of Conductor Size

Single Phase Supply

The area of the conductor is calculated using the following formula:

$$A \geq \frac{2 \times I \times K \times L}{\text{Permissible Voltage Drop}} \tag{5}$$

where:

- I = Current rating in A
- L = Length of conductor (actual) in meters. Take 20 percent greater if a straight-line approach is used.

- A = Size or area of the conductor in mm^2
- K = Resistivity of the wire:
  - For copper: $0.021\,\Omega mm^2/mm$
  - For aluminum: $0.032\,\Omega mm^2/mm$
- Permissible Voltage Drop:
  - 2% for Transformer to MDB
  - 1% for SDB to Load
  - 2% for MDB to SDB

3.4.2 Three Phase Supply

For a nearly balanced three-phase supply, the area of the conductor is calculated using the following formula:

$$A \geq \frac{1.5 \times I \times K \times L}{\text{Permissible Voltage Drop}} \tag{6}$$

3.6 Transformer Rating

The apparent power rating of a 3ϕ transformer is given by=$3V_{LN}I_{PH}10^{-3}KVA$

4 LEGENDS

LEGEND	
Preview	Block Name
	13A POWER SOCKET
	16A POWER SOCKET
	1G switch
	2 way switch
	2G Switch
	3G switch
	4G SWITCH
	6G Switch
	AC socket
	ADORE LED LAMP DIM A19-11W 5K E26
	Canopy Lights(45 W)
	Consta Pro LED Strip Light(23W)
	Coreline Downlight(25W)
	Coreline Slim Downlight(13 W)
	Hercules Led Panel (26 W)
	Phillips Green LEDi(10 W)
	FLOOR DB BOX
	SUB DISTRIBUTION BOX
	Titania LED TUBE(36 W)
	two 3g Switch
	Two way Switch
	Two way Switch X2
	EXHAUST FAN
	MAIN DISTRIBUTION BOX

Figure 1: Legends

Figure 1 shows the descriptions of symbols of light, power sockets, and DB BOXES used in the AUTOCAD drawing to design the lighting and power sub-circuit systems.

5 Electrical System Design

Using the Lumens or Light Flux Method, the approximate number of luminaires required to illuminate a given area was determined. The selection of lighting fixtures and their corresponding lumen values was based on manufacturer catalogs. The standard illuminance (lux) levels for different room types were obtained from the Indian Standard Code of Practice for Interior Illumination – IS 3646 (Part 1): 1992. Room areas were measured in AutoCAD in square feet and subsequently converted to square meters using the conversion factor.$1sq.ft = 0.0929sq.meters$. Table 1,2,3 and 4 shows the steps used to calculate the no of light luminaries in each floor respectively and was converted to the nearest number according to the shape or design of the room.

Table 1: Number of luminiares calculated for Basement Floor Plan

S.N	Location/Floor	Details	Standard Lux	Required Lux(E)	Area (sq. ft)	Area (sq. meter)-(A)	Utilization factor(μ_u)	Maintainance Factor(μ_m)	Effeciency (lumen/Watt)	Wattage per bulb	Lamp Flux in Lumen (ϕ)	Light Fixture(s) Used	No of Lamp per luminairies	No of Lumenaries=$\frac{E \times A}{\phi \mu_m \times \mu_u}$	Corrected no of Lumenaries(N)	Total Watts
1	Basement Floor Plan	18 Car Parking	20-30-50	50	2923.2029	271.5655494	0.8	0.8	125	36	4500	Titania Plus T8 LED TUBE	1	4.714679677	6	216
		Other Cars	20-30-50	50	2629.6536	244.2948194	0.8	0.8	125	36	4500	Titania Plus T8 LED TUBE	1	4.241229504	5	180
		Stairs	50-100-150	150	219.886	20.4274094	0.8	0.8	100	25	2500	Coreline Downlight	1	1.915069631	2	50
		Guard Room	50-100-150	150	61.0777	5.67411833	0.8	0.8	85	13	1100	Coreline Slim Downlight	1	1.208974076	2	26
		Store	50-100-150	150	96.9213	9.00398877	0.8	0.8	85	10	600	Phillips Green Ledi	1	3.517183113	4	40
		Electrical Duct	50-100-150	50	23.5283	2.18577907	0.8	0.8	60	11	180	ADORE LED LAMP DIM A19-11W 5K E26	1	0.948688832	1	11
		Transformer Unit	100-150-200	200	128.9783	11.98208407	0.8	0.8	133	45	3985	Canopy Light	1	0.939623908	1	45
		Bike Parking	20-30-50	50	2421.3208	224.9407023	0.8	0.8	125	36	4500	Titania Plus T8 LED TUBE	1	3.905220526	4	144
		Aune Slop	20-30-50	50	442.9767	41.15253543	0.8	0.8	130	23	3000	Consta Pro LED Strip Light	1	1.07168061	1	23

Table 2: Number of luminiaries calculated for the Ground Floor Plan

S.N	Location/Floor	Details	Standard Lux	Required Lux(E)	Area (sq. ft)	Area (sq. meter)-(A)	Utilization factor(μ_u)	Maintainance Factor(μ_m)	Effeciency (lumen/Watt)	Wattage per bulb	Lamp Flux in Lumen (ϕ)	Light Fixture(s) Used	No of Lamp per luminairies	No of Lumenaries=$\frac{E \times A}{\phi \mu_m \times \mu_u}$	Corrected no of Lumenaries(N)	Total Watts
2	Ground Floor Plan	Office 1	300-500-750	400	1504.6206	139.7792537	0.8	0.8	120	26	4400	Hercules LED PANEL	1	19.85500763	19	494
		Washroom office1	50-100-150	150	59.3326	5.51199854	0.8	0.8	85	10	1400	Philips Greenledi	1	0.922767613	1	10
		Generator Room	100-150-200	200	82.2869	7.64445301	0.8	0.8	85	13	1100	Coreline Slim Downlight	1	2.171719605	3	39
		Fire staircase going room	150-200-300	150	417.8441	38.81771689	0.8	0.8	120	26	4400	Hercules LED PANEL	1	2.06770509	3	78
		Lobby	150-200-300	150	245.581	22.8144749	0.8	0.8	120	26	4400	Hercules LED PANEL	1	1.215259672	2	52
		Staircase	50-100-150	150	216.096	20.0753184	0.8	0.8	100	25	2500	Coreline Downlight	1	1.8820611	2	50
		Electrical Duct	50-100-150	50	23.4113	2.17490977	0.8	0.8	60	11	180	ADORE LED LAMP DIM A19-11W 5K E26	1	0.943971254	1	11
		Coffee Shop	150-200-300	200	1503.9722	139.7190174	0.8	0.8	120	26	4400	Hercules LED PANEL	1	9.923225666	10	260
		Fitting room 1(mango store)	50-100-150	95	12.8334	1.19222286	0.8	0.8	60	11	180	ADORE LED LAMP DIM A19-11W 5K E26	1	0.983169893	1	11
		Fitting room2(Mango store)	50-100-150	95	12.8334	1.19222286	0.8	0.8	60	11	180	ADORE LED LAMP DIM A19-11W 5K E26	1	0.983169893	1	11
		Store	50-100-150	150	380.4574	35.34449246	0.8	0.8	120	26	4400	Hercules LED PANEL	1	1.882696686	2	52
		Mango Store	150-200-300	300	1668.8144	155.0328578	0.8	0.8	120	26	4400	Hercules LED PANEL	1	16.51628456	17	442
		Staircase(Mango Store)	50-100-150	150	124.5	11.56605	0.8	0.8	100	25	2500	Coreline Downlight	1	1.084317188	2	50
		W2f5c1	150-200-300	150	24.0737	2.23644673	0.8	0.8	85	10	600	Philips Greenledi	1	0.873612004	1	10
		Fire Staircase	50-100-150	150	79.9087	7.42351823	0.8	0.8	85	13	1100	Coreline Slim Downlight	1	1.581715532	2	26

Table 3: Number of luminiaries calculated for the First Floor Plan

S.N	Location/Floor	Details	Standard Lux	Required Lux(E)	Area (sq. ft)	Area (sq. meter)-(A)	Utilization factor(μ_u)	Maintainance Factor(μ_m)	Effeciency (lumen/Watt)	Wattage per bulb	Lamp Flux in Lumen (ϕ)	Light Fixture(s) Used	No of Lamp per luminaries	No of Lumenaries= $\frac{E*A}{\phi*\mu_m*\mu_u}$	Corrected no of Lumenaries(N)	Total Watts
3	First Floor Plan	Office 3	300-500-750	400	1504.8267	139.7984004	0.8	0.8	120	26	4400	Hercules LED PANEL	1	19.85772733	21	546
		Entrance to washroom	150-200-300	200	16.671	1.5487359	0.8	0.8	85	10	600	Philips Greenledi	1	0.806633281	1	10
		washroom w1f2CO	50-100-150	200	30.9169	2.87218001	0.8	0.8	85	10	600	Philips Greenledi	1	1.495927089	2	20
		Another washroom D2V!	50-100-150	100	40.0032	3.71629728	0.8	0.8	85	10	600	Philips Greenledi	1	0.96778575	1	10
		Fire Staircase	50-100-150	150	79.9087	7.42351823	0.8	0.8	85	13	1100	Coreline Slim Downlight	1	1.581715532	2	26
		Stair	50-100-150	150	214.2128	19.90036912	0.8	0.8	100	25	2500	Coreline Downlight	1	1.865659605	2	50
		Electrical Duct	50-100-150	50	23.4113	2.17490977	0.8	0.8	60	11	180	ADORE LED LAMP DIM A19-11W 5K E26	1	0.943971254	1	11
		Office 2	300-500-750	400	1866.5884	173.4060624	0.8	0.8	120	26	4400	Hercules LED PANEL	1	24.63154295	26	676
		Drawing Room	50-100-150	150	14.9166	1.38575214	0.8	0.8	60	11	180	ADORE LED LAMP DIM A19-11W 5K E26	1	1.804364766	2	22
		Stair to go to office 2	50-100-150	150	130.8384	12.15488736	0.8	0.8	100	25	2500	Coreline Downlight	1	1.13952069	2	50
		Washroom in trapezium room	50-100-150	150	49.0769	4.55924401	0.8	0.8	85	10	600	Philips Greenledi	1	1.780954691	2	20
		Box inside trapezium area	50-100-150	150	14.0822	1.30823638	0.8	0.8	85	10	600	Philips Greenledi	1	0.511029836	1	10
		Looby type to go washroom	50-100-150	150	100.3164	9.31939356	0.8	0.8	85	13	1100	Coreline Slim Downlight	1	1.985666241	2	26
		Riser Tread	50-100-150	150	201.55	18.723995	0.8	0.8	85	13	1100	Coreline Slim Downlight		3.989487571	4	52
		Lobby	150-200-300	150	244.2639	22.69211631	0.8	0.8	120	26	4400	Hercules LED PANEL	1	1.208741991	2	52
		Fire Staircase Entrance	150-200-300	150	297.3679	27.62547791	0.8	0.8	85	13	1100	Coreline Slim Downlight	1	5.88611035	6	78
		Drawing Room 2	50-100-150	150	14.9166	1.38575214	0.8	0.8	60	11	180	ADORE LED LAMP DIM A19-11W 5K E26	1	1.804364766	2	22

Table 4: Number of luminiaries calculated for the Second Floor Plan

S.N	Location/Floor	Details	Standard Lux	Required Lux(E)	Area (sq. ft)	Area (sq. meter)-(A)	Utilization factor(μ_u)	Maintainance Factor(μ_m)	Effeciency (lumen/Watt)	Wattage per bulb	Lamp Flux in Lumen (ϕ)	Light Fixture(s) Used	No of Lamp per luminaries	No of Lumenaries= $\frac{E*A}{\phi*\mu_m*\mu_u}$	Corrected no of Lumenaries(N)	Total Watts
4	Second Floor Plan	Office 4	300-500-750	400	1328.9864	123.4628366	0.8	0.8	120	26	4400	Hercules LED PANEL	1	17.53733474	17	442
		Lobby to washroomm	50-100-150	150	17.5962	1.63468698	0.8	0.8	85	10	600	Philips Greenledi	1	0.638549602	1	10
		Washroom D2 office 4	50-100-150	150	28.2256	2.62215824	0.8	0.8	85	10	600	Philips Greenledi	1	1.024280563	1	10
		Basin room	50-100-150	150	26.0764	2.42249756	0.8	0.8	85	10	600	Philips Greenledi	1	0.946288109	1	10
		Office 5	300-500-750	400	2106.6665	195.7093179	0.8	0.8	120	26	4400	Hercules LED PANEL	1	27.79961901	31	806
		Lobby to washroom office 5	50-100-150	150	16.6709	1.54872661	0.8	0.8	85	10	600	Philips Greenledi	1	0.604971332	1	10
		Washroom D2 office 5	50-100-150	100	30.6699	2.84923371	0.8	0.8	85	10	600	Philips Greenledi	1	0.741987945	1	10
		Basin room office 5	50-100-150	100	31.5419	2.93024251	0.8	0.8	85	10	600	Philips Greenledi	1	0.763083987	1	10
		Fire Staircase Entrance	150-200-300	150	302.3332	28.08675428	0.8	0.8	85	13	1100	Coreline Slim Downlight	1	5.984393668	6	78
		Fire Staircase	50-100-150	150	79.9087	7.42351823	0.8	0.8	85	13	1100	Coreline Slim Downlight	1	1.581715532	2	26
		Lobby	150-200-300	150	241.3108	22.41777332	0.8	0.8	120	26	4400	Hercules LED PANEL	1	1.19412855	2	52
		Stair	50-100-150	150	195.0569	18.12078601	0.8	0.8	100	25	2500	Coreline Downlight	1	1.698823688	2	50
		Electrical Duct	50-100-150	50	23.8779	2.21825691	0.8	0.8	60	11	180	ADORE LED LAMP DIM A19-11W 5K E26	1	0.962785117	1	11

Table 5: Number of luminiaries calculated for the floor (3rd $to 5^{th}$)

S.N	Location/Floor	Details	Standard Lux	Required Lux(E)	Area (sq. ft)	Area (sq. meter)-(A)	Utilization factor(μ_u)	Maintainance Factor(μ_m)	Effeciency (lumen/Watt)	Wattage per bulb	Lamp Flux in Lumen (ϕ)	Light Fixture(s) Used	No of Lamp per luminairies	No of Lumenaries=$\frac{E \times A}{\phi \mu_m \mu_u}$	Corrected no of Lumenaries(N)	Total Watts
5	Third Floor To Fifth Floor Plan	Office 6	300-500-750	400	1247.5188	115.8944965	0.8	0.8	120	26	4400	Hercules LED PANEL	1	16.46228644	18	468
		Wahsfroom lobby office 6	50-100-150	150	17.5962	1.63468698	0.8	0.8	85	10	600	Philips Greenledi	1	0.638549602	1	10
		washroom	50-100-150	150	29.84	2.772136	0.8	0.8	85	10	600	Philips Greenledi	1	1.082865625	1	10
		Basin Room	50-100-150	150	28.3315	2.63199635	0.8	0.8	85	10	600	Philips Greenledi	1	1.028123574	1	10
		Office 7	300-500-750	400	2013.7808	187.0802363	0.8	0.8	120	26	4400	Hercules LED PANEL	1	26.5738972	28	728
		Washroom lobby office 7	50-100-150	150	16.1058	1.49622882	0.8	0.8	85	10	600	Philips Greenledi	1	0.584464383	1	10
		wahshroom	50-100-150	150	24.8689	2.31032081	0.8	0.8	85	10	600	Philips Greenledi	1	0.902469066	1	10
		Basin Room	50-100-150	150	30.6148	2.84411492	0.8	0.8	85	10	600	Philips Greenledi	1	1.110982391	2	20
		Office 8	300-500-750	400	1622.9298	150.7701784	0.8	0.8	120	26	4400	Hercules LED PANEL	1	21.41621853	22	572
		Washroom Lobby office 8	50-100-150	150	16.6708	1.54871732	0.8	0.8	85	10	600	Philips Greenledi	1	0.604967703	1	10
		Washroom	50-100-150	150	31.5422	2.93027038	0.8	0.8	85	10	600	Philips Greenledi	1	1.144636867	2	20
		Basin Room	50-100-150	150	28.9533	2.68976157	0.8	0.8	85	10	600	Philips Greenledi	1	1.050688113	1	10
		Stair	50-100-150	150	201.6523	18.73349867	0.8	0.8	100	25	2500	Coreline Downlight	1	1.7562655	2	50
		Fire Staircase	50-100-150	150	79.9087	7.42351823	0.8	0.8	85	13	1100	Coreline Slim Downlight	1	1.581715532	2	26
		Electrical Duct	50-100-150	50	24.8002	2.30393858	0.8	0.8	60	11	180	ADORE LED LAMP DIM A19-11W 5K E26	1	0.999973342	1	11
		Lobby	150-200-300	150	245.3815	22.79594135	0.8	0.8	120	26	4400	Hercules LED PANEL	1	1.214272444	2	52
		Fire Staircase Entrance	150-200-300	150	301.43	28.002847	0.8	0.8	85	13	1100	Coreline Slim Downlight	1	5.966515696	6	78

Table 6: Number of luminiaries calulated for the top floor

S.N	Location/Floor	Details	Standard Lux	Required Lux(E)	Area (sq. ft)	Area (sq. meter)-(A)	Utilization factor(μ_u)	Maintainance Factor(μ_m)	Effeciency (lumen/Watt)	Wattage per bulb	Lamp Flux in Lumen (ϕ)	Light Fixture(s) Used	No of Lamp per luminairies	No of Lumenaries=$\frac{E \times A}{\phi \mu_m \mu_u}$	Corrected no of Lumenaries(N)	Total Watts
6	Top Floor Plan	Office 15	300-500-750	400	2170.6791	201.6560884	0.8	0.8	120	26	4400	Hercules LED PANEL	1	28.64433074	30	780
		Washroom Lobby	50-100-150	150	16.6129	1.54333841	0.8	0.8	85	10	600	Philips Greenledi	1	0.602866566	1	10
		Washroom	50-100-150	150	28.2256	2.62215824	0.8	0.8	85	10	600	Philips Greenledi	1	1.024280563	1	10
		Basin	50-100-150	150	26.0764	2.42249756	0.8	0.8	85	10	600	Philips Greenledi	1	0.946288109	1	10
		Fire Staircase Entrance	150-200-300	150	307.8365	28.59801085	0.8	0.8	85	13	1100	Coreline Slim Downlight	1	6.093326175	6	78
		Staircase	50-100-150	150	211.3458	19.63402482	0.8	0.8	100	25	2500	Coreline Downlight	1	1.840689827	2	50
		Lobby	150-200-300	150	265.2891	24.64535739	0.8	0.8	120	26	4400	Hercules LED PANEL	1	1.312785372	2	52
		Electrical Duct	50-100-150	50	23.2949	2.16409621	0.8	0.8	60	11	180	ADORE LED LAMP DIM A19-11W 5K E26	1	0.939277869	1	11
		Fire Staircase	50-100-150	150	79.9087	7.42351823	0.8	0.8	85	13	1100	Coreline Slim Downlight	1	1.581715532	2	26

6 Design of Distribution Boxes

The three-phase (3ϕ) supply from the transformer is fed into the main distribution box, from which it is further distributed to each floor. From the floor distribution box, power cables are routed to sub-distribution boxes on each floor as required. The power demand of each floor is assessed, and either a single-phase or a three-phase supply is allocated accordingly.

As a general guideline, a single-phase supply is used for sub-distribution boxes (SDBs) with a total load below 6000W, whereas a three-phase supply is employed for loads exceeding 6000W.

As illustrated in the single-line diagram of each floor, spare miniature circuit breakers (MCBs) are incorporated in both lighting and power subcircuits to ensure phase balancing and even load distribution.

The circuit breaker rating for each distribution box (DB) is determined using the following formula:

$$Current(I) = \begin{cases} \frac{Power(W)}{230 \times power\ factor} & \text{for single-phase supply} \\ \frac{Power(W)}{\sqrt{3} \times 400 \times power\ factor} & \text{for three-phase supply} \end{cases}$$

The nearest standard circuit breaker rating is selected based on the maximum current calculated using the above equation.

To determine the load demand on each floor, the number of power circuits was first counted, with the assumption that each power circuit can handle a total load of 500W. The air conditioning (AC) units considered in this design are assumed to be 1 ton (1250W) each. The total power demand in each sub-distribution box (SDB) and floor distribution box was then computed by summing the wattage of the lighting circuits, power circuits, and AC circuits. Based on this total load, the corresponding current was calculated, and the nearest standard MCB rating was selected accordingly. The table for calculating the load demand and TP MCCB rating for each floor is shown below:

6.1 Load and TP-MCCB rating for Basement Floor Plan

Table 7: Load Calculation and TP-MCCB rating of Basement Floor Plan

FLOOR	DB_Boxes	Power circuit	POWER WATT	NO OF LIGHT	WATTAGE OF LIGHT	TOTAL POWER CIRCUIT WATT	TOTAL LIGHT CIRCUIT WATT	TOTAL WATTS	CURRENT RATING	RATING OF MCB /MCCB USED(AFTER SDB)	3 PHASELINE CURRENT OF FDB	MCCB RATING USED
Basement floor Plan	SDB1	11	500	0	0	5500	0	5500	22.41847826	32A, DP MCB	18.55919818	40A, TP MCCB
	SDB 2	4	500	2	13	2000	609	2609	10.63451087	20A, DP MCB		
				2	10							
				15	36							
				1	23							
	SDB 3	10	500	1	11	5000	106	5106	20.8125	32A, DP MCB		
				1	45							
				2	25							
	BFDB	1	500	0	0	500	0	500	2.038043478	-		
TOTAL:								13715 W				

6.2 Load and TP-MCCB rating for Ground Floor Plan

Table 8: Load Calculation and TP-MCCB rating of Ground Floor Plan

FLOOR	DB_Boxes	Power circuit	POWER WATT	AC SOCKET	AC WATT	NO OF LIGHT	WATTAGE OF LIGHT	TOTAL LIGHT CIRCUIT WATT	TOTAL POWER CIRCUIT WATT	TOTAL WATTS	Total Current	Rating of MCB/MCCB USED	3 PHASELINE CURRENT OF FDB	MCCB RATING USED
Ground Floor Plan	SDB 1	23	500	3	1250	19	26	647	15250	15897	21.51189016	32A, TP MCCB	50.11592451	63A, TPMCCB
						1	10							
						3	13							
						2	13							
						3	26							
	SDB 2	20	500	1	1250	10	26	260	11250	11510	15.57538251	20A, TP MCCB		
	SDB 3	13	500	0	1250	21	26	628	6500	7128	9.645640878	20A, TP MCCB		
						2	25							
						2	11							
						1	10							
	GFDB	5	500	0	1250	0	0	0	2500	2500	10.19021739	-		
TOTAL:										37035 W				

6.3 Load and TP-MCCB rating for First Floor Plan

Table 9: Load Calculation and TP-MCCB rating of First Floor Plan

FLOOR	DB_Boxes	Power circuit	POWER WATT	AC SOCKET	AC WATT	NO OF LIGHT	WATTAGE OF LIGHT	TOTAL LIGHT CIRCUIT WATT	TOTAL POWER CIRCUIT WATT	TOTAL WATTS	Total Current	Rating of MCB/MCCB USED	3 PHASELINE CURRENT OF FDB	MCCB RATING USED
FIRST FLOOR PLAN	SDB1	40	500	1	1250	21	26	730	21250	21980	29.74343245	32A, TP MCCB	53.31354648	63A, TPMCCB
						3	10							
						8	13							
						2	25							
		15	500	4	1250	28	26	800	12500	13300	17.99761836	20A, TP MCCB		
						2	11							
						2	25							
	SDB 3	7	500	0	1250	3	10	108	3500	3608	14.70652174	20A, DP MCB		
						6	13							
	FFDB	1	500	0	1250	1	10	10	500	510	2.078804348	-		
TOTAL:										39398 W				

6.4 Load and TP-MCCB rating for Second Floor Plan

Table 10: Load Calculation and TP-MCCB rating of Second Floor Plan

FLOOR	DB_Boxes	Power circuit	POWER WATT	AC SOCKET	AC WATT	NO OF LIGHT	WATTAGE OF LIGHT	TOTAL LIGHT CIRCUIT WATT	TOTAL POWER CIRCUIT WATT	TOTAL WATTS	Total Current	Rating of MCB/MCCB USED	3 PHASELINE CURRENT OF FDB	MCCB RATING USED
SECOND FLOOR PLAN	SDB1	41	500	0	1250	26	31	990	20500	21490	29.0803623	32A, TP MCCB	49.76409137	63A, TPMCCB
						3	10							
						6	13							
						2	13							
						2	25							
	SDB 2	21	500	3	1250	19	26	524	14250	14774	19.99224163	25A, TP MCCB		
						3	10							
	SFDB	1	500	0	1250	1	11	11	500	511	2.082880435	-		
TOTAL:										36775 W				

6.5 Load and TP-MCCB rating for Third to Fifth Floor Plan

Table 11: Load Calculation and TP-MCCB rating of Third to Fifth Floor Plan

FLOOR	DB_Boxes	Power circuit	POWER WATT	AC SOCKET	AC WATT	NO OF LIGHT	WATTAGE OF LIGHT	TOTAL LIGHT CIRCUIT WATT	TOTAL POWER CIRCUIT WATT	TOTAL WATTS	Total Current	Rating of MCB/MCCB USED	3 PHASELINE CURRENT OF FDB	MCCB RATING USED
Third to fifth floor plan	SDB1	35	500	3	1250	23	26	772	21250	22022	29.80026703	32A, TP MCCB	85.72685118	100A, TP MCCB
						2	10							
						8	13							
						2	25							
	SDB 2	38	500	2	1250	28	26	768	21500	22268	30.13315531	32A, TP MCCB		
						4	10							
	TFDB	32	500	2	1250	20	26	561	18500	19061	25.79342884	-		
						3	10							
						1	11							
									TOTAL:	63351 W				

6.6 Load and TP-MCCB rating of Top Floor Plan

Table 12: Load Calculation and TP-MCCB rating of Top Floor Plan

FLOOR	DB_Boxes	Power circuit	POWER WATT	AC SOCKET	AC WATT	NO OF LIGHT	WATTAGE OF LIGHT	TOTAL LIGHT CIRCUIT WATT	TOTAL POWER CIRCUIT WATT	TOTAL WATTS	Total Current	Rating of MCB/MCCB USED	3 PHASELINE CURRENT OF FDB	MCCB RATING USED
Top Floor plan	SDB1	49	500	4	1250	32	26	966	29500	30466	41.22672488	50A, TP MCCB	41.91685912	50A, TP MCCB
						8	13							
						3	10							
	TOPFDB	1	500	0	1250	1	10	10	500	510	2.078804348	-		
									TOTAL:	30976 W				

7 TP-MCCB rating of Main Distribution Box

Maximum Power for 2 lifts $= 2 \times 7000W$

Maximum power for boring $= 9000W$

Maximum power for water pump $= 2$ hp $= 1492W$

Total Maximum demand of all floors $= 195775W$

Thus, maximum demand of the overall building is calculated by summing the total demand of each floor distribution boxes and lifts and water pumps.

$$= 14000 + 90000 + 1492 + 13715 + 37035 + 39398 + 36775 + 63351 + 30976$$

$$= 245,742W \approx 245,750W$$

Maximum Current $(I) =$

$$\frac{245.750}{\sqrt{3} \times 400 \times 0.8}$$

$$= 299.3964$$

So, the rating of TPMCCB used $= 300A$

7.1 Design Of Transformer

Calculation from above section shows that the total maximum demand of the building will be $P = 245,742W$which is the active power of the building. The apparent power of the transformer is given by

$$S = \frac{P}{\cos(\phi)}$$

where $cos(\phi)$ is the power factor of the building. Taking cos(ϕ)=0.8,

$$S = \frac{245750}{0.8} = 307187.5 \text{ VA} = 307.1875 \text{ kVA} \approx 310 \text{ kVA}$$

Hence, a transformer rated at 310 kVA can be used.

The full load current of the transformer is given by:

$$I = \frac{S}{\sqrt{3}V_L} = \frac{310}{\sqrt{3} \times 400} = 0.447446KA = 447.446A$$

Hence, we must use a TP-MCCB rated 500A after the transformer to protect the overall system.

8 Discussion and Conclusion

This project focuses on designing an efficient illumination system for a commercial complex, encompassing lighting design, distribution box configuration, wire sizing, and transformer selection. The use of Auto CAD ensures precision and practicality in real-world applications.

The Lumen method was employed to develop a well-structured lighting plan, involving the selection of appropriate fixtures, strategic placement for optimal illumination, and energy-efficient solutions. The distribution box was designed to facilitate effective power distribution while meeting load demands, complying with safety regulations, and minimizing energy losses.

Wire sizing calculations were performed using data from the Pioneer Cable catalog, ensuring safe and reliable power distribution while preventing issues such as overheating and voltage drops. A thorough power demand analysis and consideration of future expansion needs guided the selection of transformers, ensuring stable voltage supply across the system. Detailed technical drawings and schematics were created to enhance clarity, visualization, and documentation accuracy.

To ensure system safety and reliability, a comprehensive protection scheme was implemented, incorporating circuit breakers, fuses, and other protective devices to mitigate faults and prevent overloads. Additionally, an earthing system was integrated to effectively dissipate fault currents, reducing the risk of electrical hazards such as equipment damage and electric shocks.

In conclusion, the successful execution of this project demonstrates the capability to design and implement robust illumination systems for commercial applications. By integrating engineering principles with advanced design tools such as Auto CAD, the project delivers a lighting solution that is safe, efficient, and aesthetically optimized, tailored to meet the specific requirements of commercial environments.

9 Appendix

9.1 Luminaries Catalogue of Lights used in this project

Item Code	Product Description	System Power (W)	Power Factor	Colour Temp. (CCT)	CRI (Ra)	System Lumens (lm)	Lamp Holder
LOLDDHBND5R9	ADORE LED LAMP DIM A19-9W 3K E26	9 W	≥0.70	3000 K	≥80	800 lm	E 26
LOLDDHBNH5R9	ADORE LED LAMP DIM A19-9W 4K E26	9 W	≥0.70	4000 K	≥80	800 lm	E 26
LOLDDHBNJ5R9	ADORE LED LAMP DIM A19-9W 5K E26	9 W	≥0.70	6500 K	≥80	800 lm	E 26
LOLDDHBND5R11	ADORE LED LAMP DIM A19-11W 3K E26	11 W	≥0.70	3000 K	≥80	1100 lm	E 26
LOLDDHBNH5R11	ADORE LED LAMP DIM A19-11W 4K E26	11 W	≥0.70	4000 K	≥80	1100 lm	E 26
LOLDDHBNJ5R11	ADORE LED LAMP DIM A19-11W 5K E26	11 W	≥0.70	6500 K	≥80	1100 lm	E 26
LOLDDHBND5R15	ADORE LED LAMP DIM A19-15W 3K E26	15 W	≥0.70	3000 K	≥80	1500 lm	E 26
LOLDDHBNH5R15	ADORE LED LAMP DIM A19-15W 4K E26	15 W	≥0.70	4000 K	≥80	1500 lm	E 26
LOLDDHBNJ5R15	ADORE LED LAMP DIM A19-15W 5K E26	15 W	≥0.70	6500 K	≥80	1500 lm	E 26

Figure 2: Adore LED lamp

Figure 2 shows the 11 W bulb used in small rooms like electrical and fitting rooms.

Canopy Light

SAP Code / SAP Description	System Power (W)	Input Voltage	Lumen Efficacy	Color Temp. (CCT)	Dimming	Beam Angle
KLSN-CP3B-45WSASA2-WH50	45	100-277 VAC	> 133 lm/W	5000 K	1 -10V dimming	160 Deg.
KLSN-CP3B-55WSASA2-WH50	55	100-277 VAC	> 133 lm/W	4000k, 5000k	1 -10V dimming	160 Deg.
KLSN-CP3B-75WSASA2-WH50	75	100-277 VAC	> 133 lm/W	4000k, 5000k	1 -10V dimming	160 Deg.
KLSN-CP3B-120WSASA2-WH50	120	100-277 VAC	> 130 lm/W	5000 K	1 -10V dimming	120 Deg.
KLSN-CP3B-150WSAMA2-WH50	150	100-277 VAC	> 140 lm/W	5000 K	1 -10V dimming	110 Deg.

Figure 3: Canopy Light

Figure 3 shows the canopy lights. This light is only used in the transformer unit room of the building.

Consta Pro LED Strip Light

SAP Description SAP Code	System Power (W)	Power Factor	THD (%)	CRI (Ra)	Highest Wattage Lumen (lm) 5000 K / 4000 K / 3500 K	LPW
CONSTA PRO LED BATTEN 2FT 23W SWSC IP20 LOEDAZPNPW1W023	23 W / 18 W / 12 W	0.90	≤10	≥80	3000 lm / 3000 lm / 2800 lm	130
CONSTA PRO LED BATTEN 4FT 46W SWSC IP20 LOEDAZPNPW1W046	46 W / 35 W / 23 W	0.90	≤10	≥80	6000 lm / 6000 lm / 5700 lm	130
CONSTA PRO LED BATTEN 8FT 80W SWSC IP20 LOEDAZPNPW1W080	80 W / 60 W / 40 W	0.90	≤10	≥80	10400 lm / 10400 lm / 10000 lm	130

Nominal Voltage (V) : 120 V-277 V
Weight : 1.6 kg for 2ft | 2.8 kg for 4 ft | 4.3 kg for 8ft
Colour Fixture : White
Dimmable : 0-10V
EM Pack : Compatible
Accessories : Hanging Chain

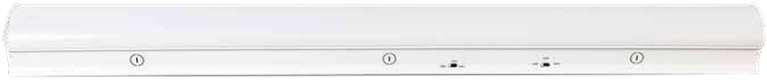


Figure 4: Consta PRO LED STRIP LIGHT

Figure 4 shows the Consta Pro LED Strip Lights. This light is used in the parking lot entrance.

Hercules LED Panel

SAP Description SAP Code	System Power (W)	Power Factor	CRI (Ra)	Highest Wattage Lumen (lm) 5000 K / 4000 K / 3500 K	Lumen Efficacy	Weight (kg)
HERCULESGEN2US2X4PLR50WLED LOEWAMPNPW1W050	50 W / 40 W / 30 W	0.90	≥80	6000 lm / 6000 lm / 5800 lm	120	3.6 kg
HERCULESGEN2US4X1PLR40WLED LOEWAOPNPW1W040	36 W / 26 W / 16 W	0.90	≥80	4400 lm / 4400 lm / 4200 lm	120	2.0 kg
HERCULESGEN2US2X2PLR40WLED LOEWANPNPW1W040	36 W / 26 W / 16 W	0.90	≥80	4400 lm / 4400 lm / 4200 lm	120	2.0 kg

Nominal Voltage (V) : 120 V-277 V
Colour Fixture : White
Dimmable : 0-10V
EM Pack : Compatible

Figure 5: Hercules LED Panel

Figure 5 shows the Hercules LED PANEL(Havells). This light is used in offices, Lobby and most parts of the building.

CoreLine Downlight www.philips.com/interact-ready

= Dimmable
= Non-Dimmable
L70 at 50.000
1 PACK

Product Type	LED	Lumen Output	Efficiency	Beam Angle	Dimmable	Color Temp.	Prod./MCB (16a type B)	IP rating	12NC	EOC code
	W	lm	lm/W	°		K	Number			
DN140B LED20S/830 C IA1	22	2200	100	120	●	3000	24	IP20	911401634405	32614600
DN140B LED20S/840 C IA1	22	2200	100	120	●	4000	24	IP20	911401634505	32615300
DN140B LED20S/840 WR IA1	22	2200	100	120	●	4000	24	IP20	911401634305	32613900
DN140B LED20S/830 C ELB3 IA1	25	2500	100	120	●	3000	24	IP20	911401640405	32841600
DN140B LED20S/830 WR ELB3 IA1	25	2200	88	120	●	3000	24	IP20	911401640205	32839300
DN140B LED20S/840 C ELB3 IA1	25	2200	88	120	●	4000	24	IP20	911401640505	32842300
DN140B LED20S/840 WR ELB3 IA1	25	2200	88	120	●	4000	24	IP20	911401640305	32840900

Figure 6: Coreline Downlight

Figure 6 shows the Phillips coreline downlight model. It is used in stairs of the building.

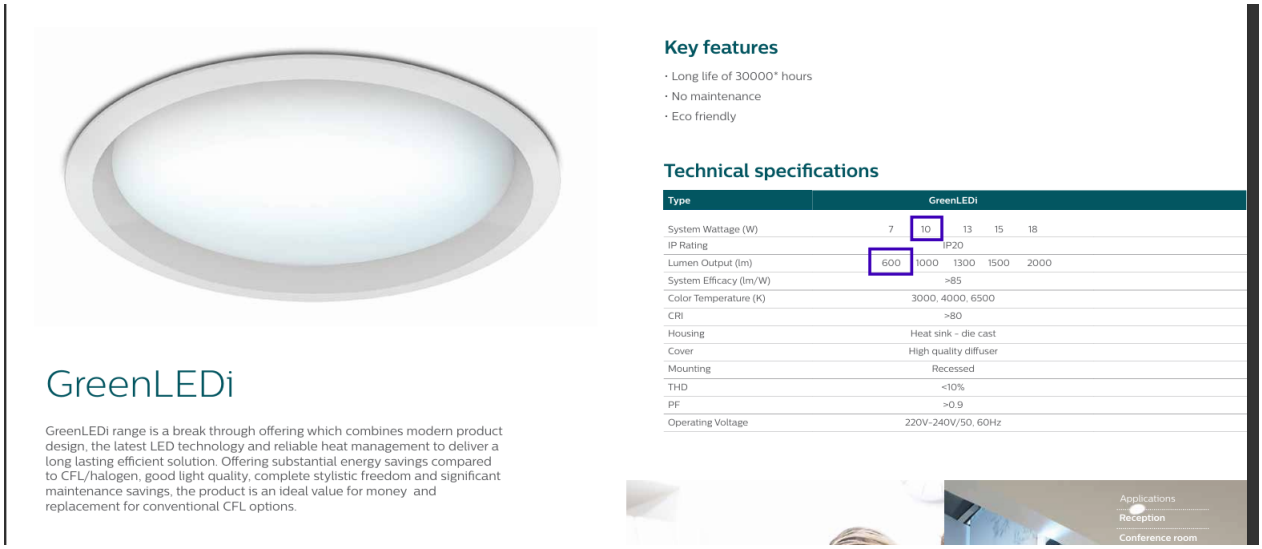
CoreLine SlimDownlight www.philips.com/coreline

= Dimmable
= Non-Dimmable
L70 at 50.000
10/16 PACK

Product Type	LED	Lumen Output	Efficiency	Dimmable	Color Temp.	Prod./MCB (16a type B)	IP rating	12NC	EOC code
	W	lm	lm/W		K	Number			
1 DN145B LED6S/830 PSU II WH	6.5	650	100	✕	3000	27	IP20/44	911401805780	33945099
1 DN145B LED6S/840 PSU II WH	6.5	650	100	✕	4000	27	IP20/44	911401805880	33946799
2 DN145B LED10S/830 PSU II WH	11	1100	100	✕	3000	38	IP20/44	911401805980	33947499
2 DN145B LED10S/840 PSU II WH	11	1100	100	✕	4000	38	IP20/44	911401806080	33948199
3 DN145C LED10S/830 PSU II WH	11	1100	100	✕	3000	38	IP44	911401806380	33951199
3 DN145C LED10S/840 PSU II WH	11	1100	100	✕	4000	38	IP44	911401806480	33952899
2 DN145B LED10S/830 PSD-E II WH	13	1100	85	●	3000	24	IP20/44	910500465901	96100899
2 DN145B LED10S/840 PSD-E II WH	13	1100	85	●	4000	24	IP20/44	910500465902	96101599
4 DN145B LED20S/830 PSU II WH	21	2100	100	✕	3000	13	IP20/44	911401806180	33949899
4 DN145B LED20S/840 PSU II WH	21	2100	100	✕	4000	13	IP20/44	911401806280	33950499
5 DN145C LED20S/830 PSU II WH	21	2100	100	✕	3000	13	IP44	911401806580	33953599
5 DN145C LED20S/840 PSU II WH	21	2100	100	✕	4000	13	IP44	911401806680	33954299
4 DN145B LED20S/830 PSD-E II WH	22.5	2100	93	●	3000	24	IP20/44	910500465903	96102299
4 DN145B LED20S/840 PSD-E II WH	22.5	2100	93	●	4000	24	IP20/44	910500465904	96103999
6 DN135B LED20S/830 PSD-E II WH	28	2000	71	●	3000	24	IP44	910500465817	96492499
6 DN135B LED20S/840 PSD-E II WH	28	2000	71	●	4000	24	IP44	910500465818	96493199

Figure 7: Coreline Slim Downlight

Figure 7shows the phillips coreline slim downlight. It is used in corridors towards the emergency exit and in the generator room of the building.



GreenLEDi

GreenLEDi range is a break through offering which combines modern product design, the latest LED technology and reliable heat management to deliver a long lasting efficient solution. Offering substantial energy savings compared to CFL/halogen, good light quality, complete stylistic freedom and significant maintenance savings, the product is an ideal value for money and replacement for conventional CFL options.

Key features

- Long life of 30000* hours
- No maintenance
- Eco friendly

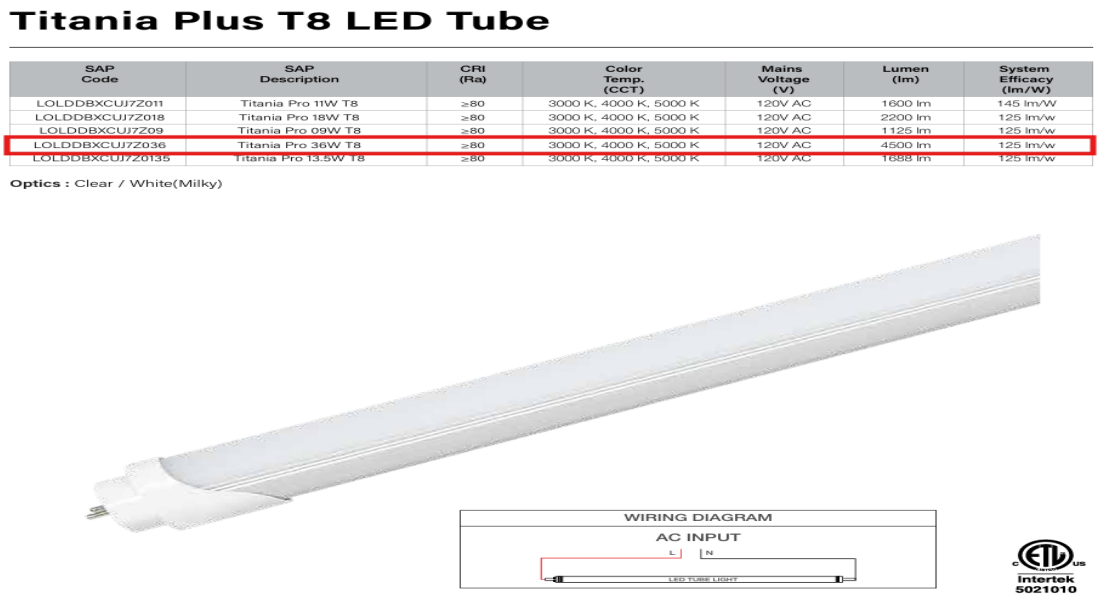
Technical specifications

Type	GreenLEDi				
System Wattage (W)	7	10	13	15	18
IP Rating	IP20				
Lumen Output (lm)	600	1000	1300	1500	2000
System Efficacy (lm/W)	>85				
Color Temperature (K)	3000, 4000, 6500				
CRI	>80				
Housing	Heat sink - die cast				
Cover	High quality diffuser				
Mounting	Recessed				
THD	<10%				
PF	>0.9				
Operating Voltage	220V-240V/50, 60Hz				

Applications
Reception
Conference room

Figure 8: phillips greeen LEDi

Figure 8 shows the 10 W Bulb. It is used throughout the washrooms in the buildings. This light is used to illuminate the parking lot.



Titania Plus T8 LED Tube

SAP Code	SAP Description	CRI (Ra)	Color Temp. (CCT)	Mains Voltage (V)	Lumen (lm)	System Efficacy (lm/W)
LOLDDBXCUJ7Z011	Titania Pro 11W T8	≥80	3000 K, 4000 K, 5000 K	120V AC	1600 lm	145 lm/W
LOLDDBXCUJ7Z018	Titania Pro 18W T8	≥80	3000 K, 4000 K, 5000 K	120V AC	2200 lm	125 lm/w
LOLDDBXCUJ7Z09	Titania Pro 09W T8	≥80	3000 K, 4000 K, 5000 K	120V AC	1125 lm	125 lm/w
LOLDDBXCUJ7Z036	Titania Pro 36W T8	≥80	3000 K, 4000 K, 5000 K	120V AC	4500 lm	125 lm/w
LOLDDBXCUJ7Z0135	Titania Pro 13.5W T8	≥80	3000 K, 4000 K, 5000 K	120V AC	1688 lm	125 lm/w

Optics : Clear / White(Milky)

Figure 9: TITANIA PLUS T8 LED TUBE.png

9.2 Cable Catalogue

- For 16 A power sub socket, copper wire of dimension $2 \times 4 + 1 \times 2.5\,\text{mm}^2$ cable is used.
- For 13 A power sub socket, copper wire of dimension $2 \times 2.5 + 1 \times 2.5\,\text{mm}^2$ cable is used and for lighting circuit, copper wire of dimension $2 \times 1.5\,\text{mm}^2$ cable is used.
- The 4-core PVC insulated armoured cable is used for high current applications, i.e., for MDB to FDB and Transformer to MDB.
- In case of supply from FDB to SDB with single-phase supply, 2-core PVC insulated unarmoured cable is used.
- For current rating < 63 A, copper wire is used, while for current rating greater than that, aluminum wire is used to make the system economical.
- For Earthing Conductor, Copper Strip of $25mm\ X\ 3mm$ is used in Lightning Arrester, Body of the transformer and its winding and the main distribution box.

The following ratings cable are choosen:

TABLE 16: 2 CORE PVC INSULATED ARMOURED & UNARMOURED CABLE WITH ALUMINIUM/ COPPER CONDUCTOR (AS-IS:1554(1))

Area	Cond.Min. No. of wires		Insulation Thick-ness	Inner Thick-ness	Dimension of Armour Wire Strip	Outer Thickness		Approx O.D.		Approx. Net Wt.Of Cable				Max. D.C. resistance at 20°C		Approx. A.C. resistance at operating temp. 70°C		Approx capacitance/phase		Current Rating						Short Circuit rating for 1 sec.	
						Arm	Un-arm	Arm	Un-arm	Arm		Un-arm						Arm	Un-arm	In ground		In Duct		In air			
mm²	No.		mm	mm	mm	mm	mm	mm	mm	Kg/km		Kg/km		Ohm/km		Ohm/km		µf/ km		Amps		Amps		Amps		KA(rms)	
Al/Cu	Al	Cu	Al/Cu	Al/Cu	Al/Cu	Al/Cu	Al/Cu	Al/Cu	Al/cu	Al	Cu	Al	Cu	Al	Cu	Al	Cu	Al/Cu	Al/Cu	Al	Cu	Al	Cu	Al	Cu	Al	Cu
2.5	1.0	1.0	0.9	0.3	1.4	1.24	1.8	14.8	13.0	436	477	199	230	12.1	7.41	14.5	8.87	0.13	0.13	25	32	21	27	21	27	0.19	0.288
4	1.0	1.0	1.0	0.3	1.4	1.24	1.8	16.1	14.3	502	554	242	294	7.41	4.61	8.9	5.52	0.14	0.14	32	41	27	35	27	35	0.304	0.46
6	1.0	1.0	1.0	0.3	1.4	1.24	1.8	18.1	16.3	633	707	316	390	4.61	3.08	5.54	3.69	0.16	0.16	40	50	34	44	35	45	0.456	0.69
10	1.0	7.0	1.0	0.3	1.4	1.24	1.8	19.5	17.7	734	858	384	508	3.08	1.83	3.70	2.19	0.18	0.18	55	70	45	58	47	60	0.760	1.15
16	7.0	7.0	1.0	0.3	4 x 0.8	1.4	1.8	17.5	16.4	533	729	325	521	1.91	1.15	2.30	1.38	0.19	0.19	70	90	58	75	59	78	1.22	1.84
25	7.0	7.0	1.2	0.3	4 x 0.8	1.4	2.0	20.1	19.4	687	991	453	757	1.20	0.727	1.44	0.87	0.22	0.22	90	115	76	97	78	105	1.90	2.88
35	7.0	7.0	1.2	0.3	4 x 0.8	1.4	2.0	21.5	20.8	796	1226	538	972	0.868	0.524	1.04	0.627	0.24	0.24	110	140	92	120	99	125	2.66	4.03
50	7.0	7.0	1.4	0.3	4 x 0.8	1.4	2.0	24.1	23.4	976	1549	676	1249	0.641	0.387	0.77	0.463	0.24	0.24	135	165	115	145	125	155	3.80	5.75
70	19	19	1.4	0.3	4 x 0.8	1.56	2.0	26.7	25.6	1207	2072	842	1707	0.443	0.268	0.532	0.321	0.26	0.26	160	205	140	180	150	195	5.32	8.05
95	19	19	1.6	0.4	4 x 0.8	1.56	2.2	30.1	29.4	1501	2668	1122	2289	0.32	0.193	0.384	0.231	0.26	0.26	190	240	170	215	185	230	7.22	10.09
120	19	19	1.6	0.4	4 x 0.8	1.56	2.2	32.1	31.4	1743	3209	1309	2775	0.253	0.153	0.304	0.184	0.28	0.28	210	275	190	235	210	265	9.12	13.8
150	19	19	1.8	0.4	4 x 0.8	1.72	2.4	34.7	34	2053	3906	1577	3430	0.206	0.124	0.248	0.149	0.28	0.28	240	310	210	270	240	305	11.4	17.3
185	37	37	2.0	0.5	4 x 0.8	1.88	2.4	38	37	2461	4744	1914	4197	0.164	0.0991	0.198	0.120	0.28	0.28	275	350	240	300	275	350	14.1	21.3
240	37	37	2.2	0.5	4 x 0.8	2.04	2.6	43	42	3034	5891	2425	5282	0.125	0.0754	0.152	0.0912	0.28	0.28	320	405	275	345	325	410	18.2	27.6
300	37	37	2.4	0.6	4 x 0.8	2.20	2.8	46.5	45.6	3636	7297	2964	6625	0.100	0.0601	0.122	0.0739	0.29	0.29	355	430	305	385	365	465	22.8	34.5
400	61	61	2.6	0.7	4 x 0.8	2.36	3.2	52.3	51.8	4495	9461	3782	8748	0.0778	0.0470	0.096	0.0592	0.29	0.29	385	490	345	425	420	530	30.4	46.0
500	61	61	3.0	0.7	4 x 0.8	2.84	3.4	58.7	57.6	5419	9558	4732	8871	0.0605	0.0366	0.076	0.0468	0.29	0.29	415	520	365	460	455	575	38.0	57.5
630	61	61	3.4	0.7	4 x 0.8	2.84	3.8	64.8	64.4	6875	12247	6012	11384	0.0469	0.0283	0.061	0.0379	0.29	0.29	460	565	405	510	520	655	47.9	72.5

Figure 10: 2 CORE PVC INSULATED UNARMOURED CABLE

Area	Cond. Min. No. of wires		Insulation Thickness	Inner Thick-ness	Dimension of Armour Wire Strip	Outer Thickness		Approx. O.D.		Approx. Net Wt. Of Cable				Max. D.C. resistance at 20°C		Approx. A.C. resistance at operating temp. 70°C		Approx capacitance/phase		Current Rating						Short Circuit rating for 1 sec.	
						Arm	Un-Arm	Arm	Un-arm	Arm		Un-arm						Arm	Un-arm	In ground		In Duct		In air			
mm²	No.		mm	mm	mm	mm	mm	Mm	mm	Kg/km		Kg/km		Ohm/km		Ohm/km		µf/ km		Amps		Amps		Amps		KA (rms)	
Al/Cu	Al	cu	Al/Cu	Al/Cu	Al/Cu	Al/Cu	Al/Cu	Al/Cu	Al/cu	Al	Cu	Al	Cu	Al	Cu	Al	Cu	Al/Cu	Al/Cu	Al	Cu	Al	Cu	Al	Cu	Al	Cu
2.5	1	1	0.9	0.3	1.4	1.24	1.8	16.5	14.7	552	614	253	335	12.1	7.41	14.5	8.87	0.355	0.355	21	27	18	24	18	24	0.19	0.288
4	1	1	1.0	0.3	1.4	1.24	1.8	17.6	15.7	638	735	320	417	7.41	4.61	8.9	5.52	0.395	0.395	28	36	23	30	23	30	0.3	0.46
6	1	1	1.0	0.3	1.4	1.24	1.8	18	17.1	792	941	419	568	4.61	3.08	5.54	3.69	0.435	0.435	35	45	30	38	30	39	0.46	0.69
10	1	7	1.0	0.3	4 x 0.8	1.4	1.8	21.7	20.5	824	1058	517	767	3.08	1.83	3.70	2.19	0.495	0.495	46	60	39	50	40	52	0.76	1.15
16	7	7	1.0	0.3	4 x 0.8	1.4	2.0	22.1	21.4	822	1214	555	947	1.91	1.15	2.30	1.38	0.56	0.56	60	77	50	64	51	66	1.22	1.84
25	7	7	1.2	0.3	4 x 0.8	1.4	2.0	25.3	25	1060	1668	748	1356	1.20	0.727	1.44	0.87	0.62	0.62	76	99	63	81	70	90	1.90	2.88
35	7	7	1.2	0.3	4 x 0.8	1.4	2.0	27.6	26.9	1261	2129	907	1775	0.868	0.524	1.04	0.627	0.66	0.66	92	120	77	99	86	110	2.66	4.03
50	7	7	1.4	0.4	4 x 0.8	1.56	2.2	33.0	31	1593	2740	1215	2362	0.641	0.387	0.77	0.463	0.7	0.7	110	145	95	125	105	135	3.80	5.75
70	19	19	1.4	0.4	4 x 0.8	1.56	2.2	35.1	34.4	1910	3711	1530	3259	0.443	0.268	0.532	0.321	1.16	1.16	135	175	115	150	130	165	5.32	8.05
95	19	19	1.6	0.4	4 x 0.8	1.72	2.4	39.4	38.7	2527	4859	2012	4344	0.32	0.193	0.384	0.231	1.18	1.18	165	210	140	175	155	200	7.22	10.9
120	19	19	1.6	0.5	4 x 0.8	1.88	2.4	43.1	42.1	3004	5935	2400	5331	0.253	0.153	0.304	0.184	1.31	1.31	185	240	155	195	180	230	9.12	13.8
150	19	19	1.8	0.5	4 x 0.8	1.88	2.6	47	46.4	3538	7245	2913	6620	0.206	0.124	0.248	0.149	1.28	1.28	210	270	175	225	205	265	11.4	17.3
185	37	37	2.0	0.6	4 x 0.8	2.04	2.8	52.1	51.5	4289	8856	3609	8175	0.164	0.0991	0.198	0.120	1.3	1.3	235	300	200	255	240	305	14.1	21.3
240	37	37	2.2	0.6	4 x 0.8	2.36	3.0	59	58.1	5394	11107	4573	10286	0.125	0.0754	0.152	0.0912	1.34	1.34	275	345	235	295	280	355	18.2	27.6
300	37	37	2.4	0.7	4 x 0.8	2.52	3.4	65.8	65.7	6553	13927	5755	13129	0.100	0.0601	0.122	0.0739	1.37	1.37	305	385	260	335	315	400	22.8	34.5
400	61	61	2.6	0.7	4 x 0.8	2.68	3.6	73.3	73.1	8080	18011	7205	17136	0.0778	0.0470	0.096	0.0592	1.43	1.43	335	425	290	360	375	455	30.4	46.0
500	61	61	3.0	0.7	4 x 0.8	3.0	4.0	83.1	83	10117	22483	9150	21516	0.0605	0.0366	0.076	0.0468	1.41	1.41	370	440	320	390	540	500	37.9	57.5

Figure 11: 4 CORE PVC INSULATED UNARMOURED CABLE

TABLE 19: 4 CORE PVC INSULATED ARMOURED & UNARMOURED CABLE WITH ALUMINIUM/ COPPER CONDUCTOR. (AS-IS:1554(1))

Area	Cond. Min. No. of wires		Insulation Thickness	Inner Thick-ness	Dimension of Armour Wire Strip	Outer Thickness		Approx. O.D.		Approx. Net Wt. Of Cable				Max. D.C. resistance at 20°C		Approx. A.C. resistance at operating temp. 70°C		Approx capacitance/phase		Current Rating						Short Circuit rating for 1 sec.	
						Arm	Un-Arm	Arm	Un-arm	Arm		Un-arm						Arm	Un-arm	In ground		In Duct		In air			
mm²	No.		mm	mm	mm	mm	mm	Mm	mm	Kg/km		Kg/km		Ohm/km		Ohm/km		µf/ km		Amps		Amps		Amps		KA (rms)	
Al/Cu	Al	cu	Al/Cu	Al/Cu	Al/Cu	Al/Cu	Al/Cu	Al/Cu	Al/cu	Al	Cu	Al	Cu	Al	Cu	Al	Cu	Al/Cu	Al/Cu	Al	Cu	Al	Cu	Al	Cu	Al	Cu
2.5	1	1	0.9	0.3	1.4	1.24	1.8	16.5	14.7	552	614	253	335	12.1	7.41	14.5	8.87	0.355	0.355	21	27	18	24	18	24	0.19	0.288
4	1	1	1.0	0.3	1.4	1.24	1.8	17.6	15.7	638	735	320	417	7.41	4.61	8.9	5.52	0.395	0.395	28	36	23	30	23	30	0.3	0.46
6	1	1	1.0	0.3	1.4	1.24	1.8	18	17.1	792	941	419	568	4.61	3.08	5.54	3.69	0.435	0.435	35	45	30	38	30	39	0.46	0.69
10	1	7	1.0	0.3	4 x 0.8	1.4	1.8	21.7	20.5	824	1058	517	767	3.08	1.83	3.70	2.19	0.495	0.495	46	60	39	50	40	52	0.76	1.15
16	7	7	1.0	0.3	4 x 0.8	1.4	2.0	22.1	21.4	822	1214	555	947	1.91	1.15	2.30	1.38	0.56	0.56	60	77	50	64	51	66	1.22	1.84
25	7	7	1.2	0.3	4 x 0.8	1.4	2.0	25.3	25	1060	1668	748	1356	1.20	0.727	1.44	0.87	0.62	0.62	76	99	63	81	70	90	1.90	2.88
35	7	7	1.2	0.3	4 x 0.8	1.4	2.0	27.6	26.9	1261	2129	907	1775	0.868	0.524	1.04	0.627	0.66	0.66	92	120	77	99	86	110	2.66	4.03
50	7	7	1.4	0.4	4 x 0.8	1.56	2.2	33.0	31	1593	2740	1215	2362	0.641	0.387	0.77	0.463	0.7	0.7	110	145	95	125	105	135	3.80	5.75
70	19	19	1.4	0.4	4 x 0.8	1.56	2.2	35.1	34.4	1910	3711	1530	3259	0.443	0.268	0.532	0.321	1.16	1.16	135	175	115	150	130	165	5.32	8.05
95	19	19	1.6	0.4	4 x 0.8	1.72	2.4	39.4	38.7	2527	4859	2012	4344	0.32	0.193	0.384	0.231	1.18	1.18	165	210	140	175	155	200	7.22	10.9
120	19	19	1.6	0.5	4 x 0.8	1.88	2.4	43.1	42.1	3004	5935	2400	5331	0.253	0.153	0.304	0.184	1.31	1.31	185	240	155	195	180	230	9.12	13.8
150	19	19	1.8	0.5	4 x 0.8	1.88	2.6	47	46.4	3538	7245	2913	6620	0.206	0.124	0.248	0.149	1.28	1.28	210	270	175	225	205	265	11.4	17.3
185	37	37	2.0	0.6	4 x 0.8	2.04	2.8	52.1	51.5	4289	8856	3609	8175	0.164	0.0991	0.198	0.120	1.3	1.3	235	300	200	255	240	305	14.1	21.3
240	37	37	2.2	0.6	4 x 0.8	2.36	3.0	59	58.1	5394	11107	4573	10286	0.125	0.0754	0.152	0.0912	1.34	1.34	275	345	235	295	280	355	18.2	27.6
300	37	37	2.4	0.7	4 x 0.8	2.52	3.4	65.8	65.7	6553	13927	5755	13129	0.100	0.0601	0.122	0.0739	1.37	1.37	305	385	260	335	315	400	22.8	34.5
400	61	61	2.6	0.7	4 x 0.8	2.68	3.6	73.3	73.1	8080	18011	7205	17136	0.0778	0.0470	0.096	0.0592	1.43	1.43	335	425	290	360	375	455	30.4	46.0
500	61	61	3.0	0.7	4 x 0.8	3.0	4.0	83.1	83	10117	22483	9150	21516	0.0605	0.0366	0.076	0.0468	1.41	1.41	370	440	320	390	540	500	37.9	57.5

Figure 12: Caption